

2.1

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i$ " 5	$\sum y_i$ " 10	$\sum (x_i - \bar{x})$ " 0	$\sum (x_i - \bar{x})^2$ " 10	$\sum (y_i - \bar{y})$ " 0	$\sum (x_i - \bar{x})(y_i - \bar{y})$ " 8

(a) $\bar{x} = 1, \bar{y} = 2$

(b)
$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{8}{10} = 0.8$$

$$b_1 = \bar{y} - b_2 \bar{x} = 2 - 0.8 \times 1 = 1.2$$

(c)

$$\sum x_i^2 = 15, \sum x_i y_i = 18$$

$$\begin{aligned} \sum (x_i - \bar{x})^2 &= 10 = \sum x_i^2 - N \bar{x}^2 \\ &= 15 - 5 \cdot 1^2 = 10 \quad \text{QED} \end{aligned}$$

$$\begin{aligned} \sum (x_i - \bar{x})(y_i - \bar{y}) &= 8 = \sum x_i y_i - N \bar{x} \bar{y} \\ &= 18 - 5 \cdot 1 \cdot 2 \\ &= 8 \quad \text{QED} \end{aligned}$$

(d)

x	y	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2	1	1	1
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i$ 5	$\sum y_i$ 10	$\sum \hat{y}_i$ 10	$\sum \hat{e}_i$ 0	$\sum \hat{e}_i^2$ 3.6	$\sum x_i \hat{e}_i$ 0

$$S_{yy}^2 = \frac{\sum (y_i - \bar{y})^2}{n-1} = \frac{10}{4} = 2.5$$

x	y	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
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$\sum x_i$ 5	$\sum y_i$ 10	$\sum \hat{y}_i$ 10	$\sum \hat{e}_i$ 0	$\sum \hat{e}_i^2$ 3.6	$\sum x_i \hat{e}_i$ 0

$$S_y^2 = \frac{\sum (y_i - \bar{y})^2}{N-1} = \frac{10}{4} = 2.5$$

$$S_x^2 = \frac{\sum (x_i - \bar{x})^2}{N-1} = \frac{10}{4} = 2.5$$

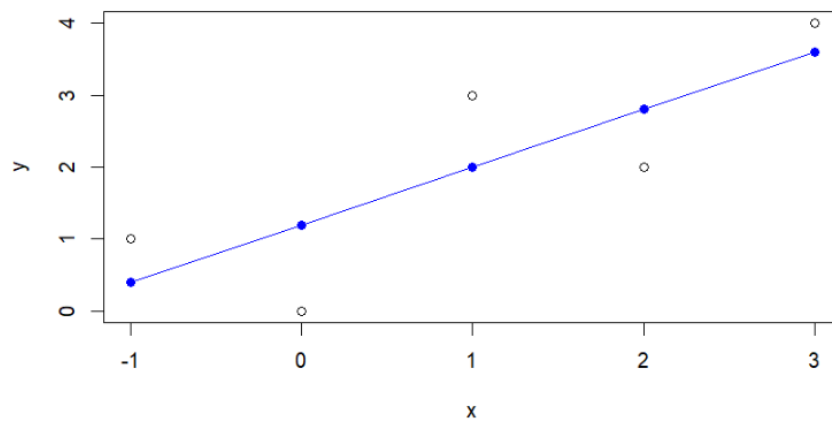
$$S_{xy} = \frac{\sum (y_i - \bar{y})(x_i - \bar{x})}{N-1} = \frac{8}{4} = 2$$

$$r_{xy} = \frac{S_{xy}}{S_x S_y} = \frac{2}{2.5} = 0.8$$

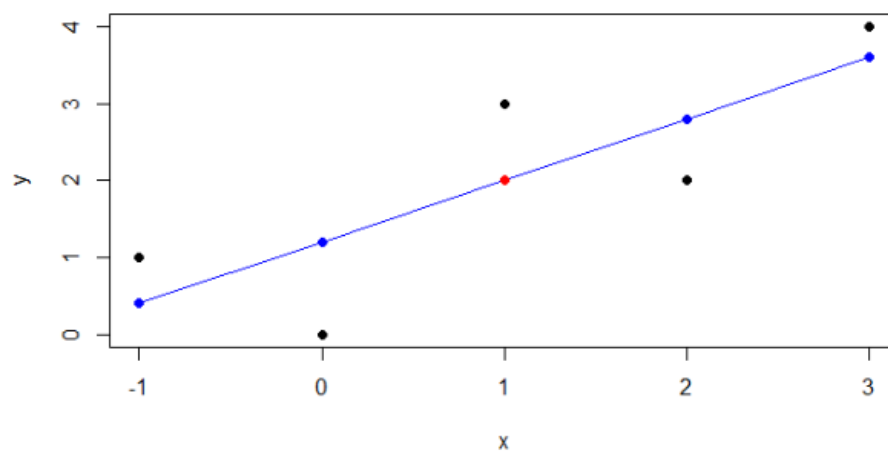
$$CV_x = 100 \cdot \left(\frac{S_x}{\bar{x}} \right) = 100 \cdot \frac{\sqrt{2.5}}{1} = 158.11$$

the median of $x = 1$

(e)



(f)



Yes, the fitted regression line passes through that point.

(g)

$$\bar{x}=1, \bar{y}=2, b_1=1.2, b_2=0.8$$
$$\Rightarrow \bar{y}=2=1.2+0.8 \times 1=b_1+b_2 \bar{x} \quad \text{QED}$$

(h)

$$\overline{\hat{y}} = \frac{\sum \hat{y}_i}{N} = \frac{10}{5} = 2 = \bar{y} \quad \text{QED}$$

(i)

$$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

(j)

$$\widehat{\text{var}}(b_2 | X) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$$

$$\text{se}(b_2) = \sqrt{0.12} = 0.3464$$

2.22

(a)

$$\text{TOTALSCORE} = 918.628 + 9.989 \text{ SMALL}$$

- 實施普通班(當 SMALL 等於 0 時)，預期平均總成績 918.628 分。
- 實施小班制的預期平均總成績會比實施普通班增加 9.989 分。
- 在 10 % 的 significance level 之下，我們有 90 % 的信心 concludes 實施小班制能夠幫助平均總成績提升。

(b)

$$\text{READSCORE} = 434.123 + 5.466 \text{ SMALL}$$

- 5.466 小於(a)的迴歸係數 9.989，實施小班制造成預期平均閱讀成績增加的幅度小於預期平均總成績增加的幅度。

$$\text{MATHSCORE} = 484.505 + 4.522 \text{ SMALL}$$

- Estimated slope 的 t-value 並不顯著，代表著實施小班制的預期平均數學成績與實施普通班並沒有顯著差異，跟上述實施小班制能夠幫助平均總成績提升的結論相異。

(c)

$$\text{TOTALSCORE} = 922.145 - 1.396 \text{ AIDE}$$

- 沒有增加助教的普通班預期平均總成績 922.145 分。
- Estimated slope 的 t-value 並不顯著，代表著增加助教與否對普通班的預期平均總成績並沒有顯著差異。

(d)

$$\text{READSCORE} = 435.9084 - 0.3719 \text{ AIDE}$$

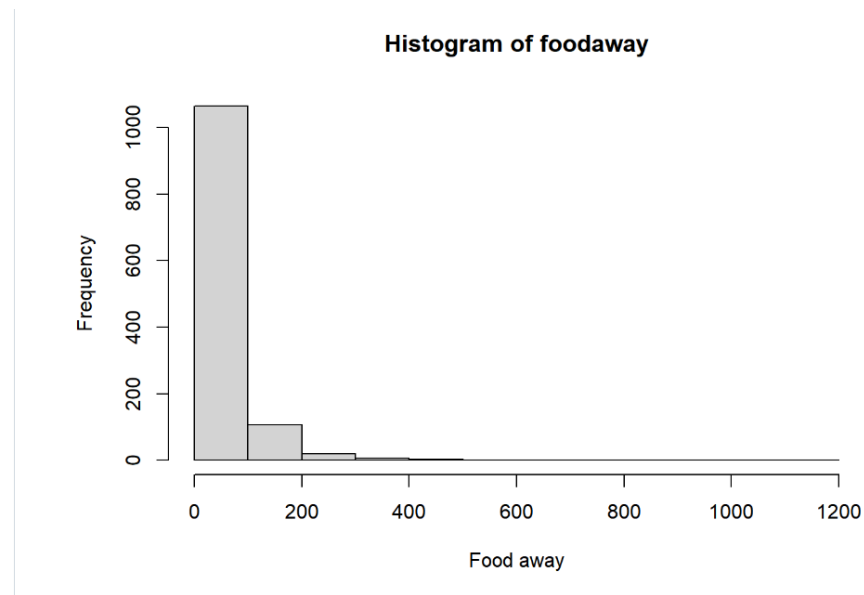
- Estimated slope 的 t-value 並不顯著，代表著增加助教與否對普通班的預期平均閱讀成績並沒有顯著差異。

$$\text{MATHSCORE} = 486.236 - 1.024 \text{ AIDE}$$

- Estimated slope 的 t-value 並不顯著，代表著增加助教與否對普通班的預期平均數學成績並沒有顯著差異。

2.25

(a)



mean = 49.27, median value = 32.56, 25th percentile = 12.04, 75th percentile = 67.60

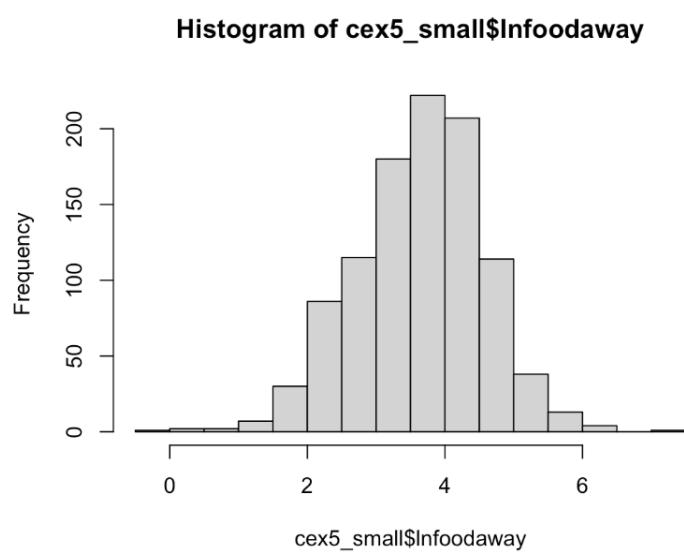
(b)

For households including a member with an advanced degree, the mean is 73.15 and median value is 48.15.

For households including a member with a college degree member, the mean is 48.60 and median value is 36.11.

For households including a member with no advanced or college degree member, the mean is 39.01 and median value is 26.02.

(c)



mean = 3.6508 , median value = 3.6865 , 25th percentile = 3.0759 , 75th percentile =

4.2797

Why FOODAWAY and $\ln(\text{FOODAWAY})$ have different numbers of observations?

- There are some 0 in the raw data of FOODAWAY, but $\ln(0) = -\infty$. Therefore, replace $-\infty$ with NA.

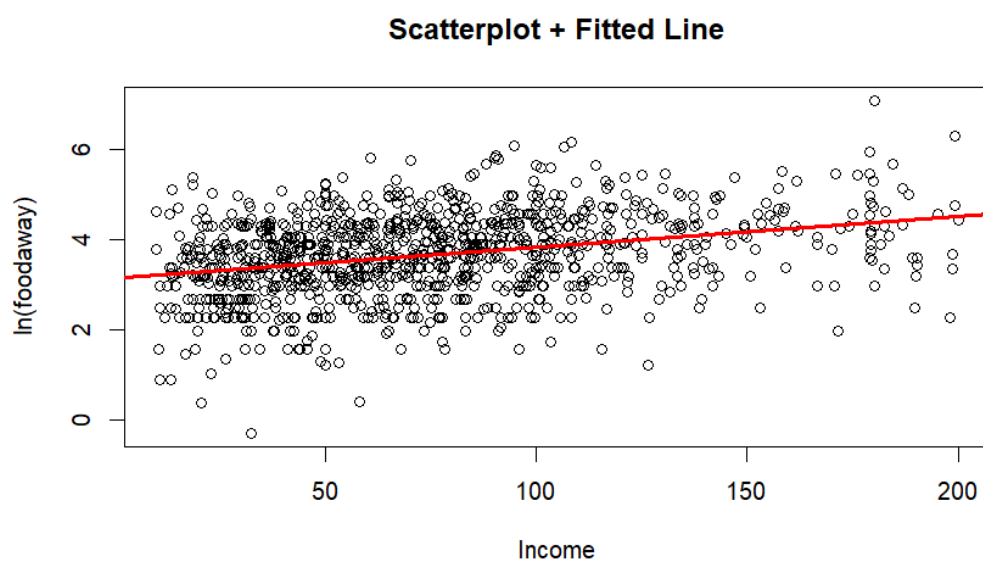
(d)

$$\ln(\text{FOODAWAY}) = 3.1293 + 0.0069 \text{ INCOME}$$

Interpretation of the estimated slope :

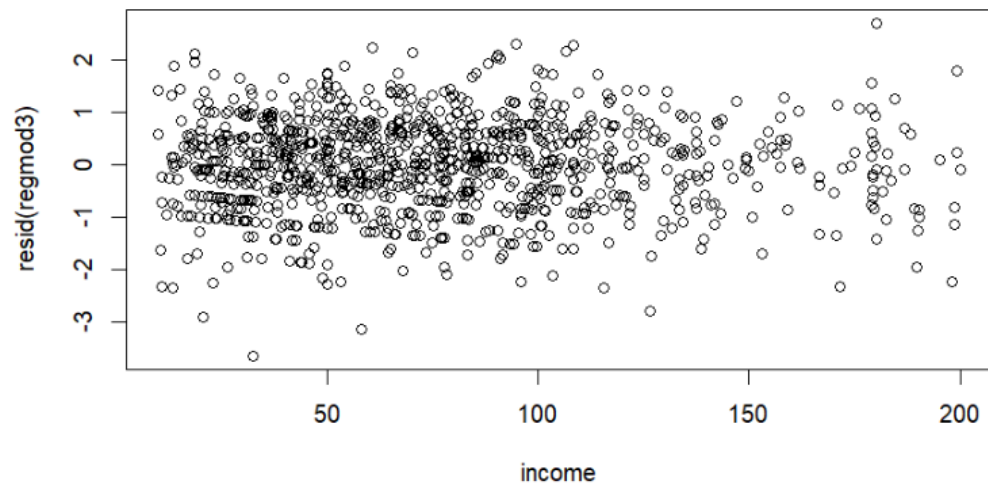
在其他因素控制不變的情況下，當 household INCOME 每增加 1 單位(100 元)時，每人的 FOODAWAY expenditures 的預期值會增加 0.69 %

(e)



(f)

	1	2	3	4	5
6	-1.210546e-01	8.031280e-01	-4.154873e-03	NA	9.967711e-01
	5.826628e-01				
	7	8	9	10	11
12	6.895088e-01	9.704880e-01	-1.706670e-03	2.864760e-01	-1.780234e+00
	1.394674e+00				
	13	14	15	16	17
18	NA	5.686162e-01	1.000108e+00	-4.928078e-01	4.009564e-01
	3.773583e-01				
	19	20	21	22	23
24	5.758635e-01	NA	NA	-1.077474e+00	2.651890e-01
	7.497477e-01				
	25	26	27	28	29
30	-1.916155e-01	-1.258769e+00	1.270226e+00	2.055229e+00	NA
	-1.071470e+00				



The patterns of the least squares residuals vs. INCOME seem completely random.